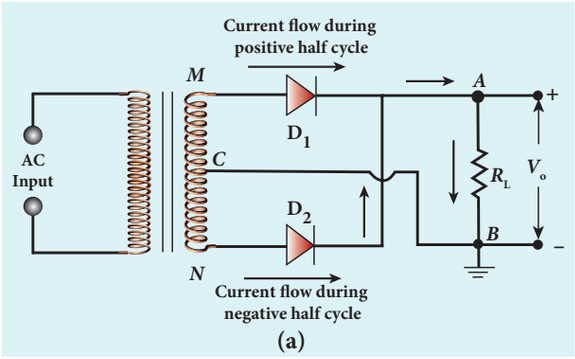


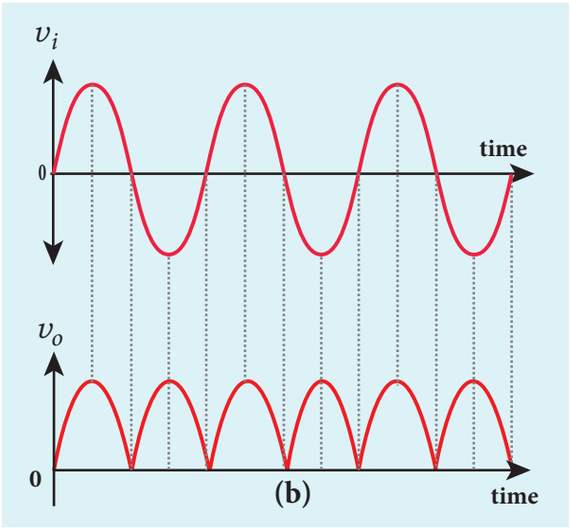


Figure 10.18 (a) Full wave rectifier circuit (b) Input and output waveforms

potential and M is at negative potential. This forward biases diode D_2 and reverse biases diode D_1. Hence, being forward biased, diode D_2 conducts and current flows along the path ND_2ABC.

During both postive and negative half cycles of the input signal, the current flows through the load in the same direction. The output signal corresponding to the input signal is shown in Figure 10.18(b). Though both half cycles of AC input are rectified, the output is still pulsating in nature.

The efficiency (η) of full wave rectifier is twice that of a half wave rectifier and is found to be 81.2 %. It is because of power losses in the winding, the diode and the load resistance.

Note

Centre tap transformer: There is a facility to tap at halfway point in the secondary windings. This helps to measure the induced voltage from one end of the secondary to the centre point. If the centre tap point is grounded, then the voltage across the secondary will be divided into two halves. For example, if the voltage across the secondary is 240 V, then the voltage across one end and the centre tap point is +120 V and at the other end it is –120 V.

10.3.5 Breakdown mechanism

The reverse current or the reverse saturation current due to the minority charge carriers is small. If the reverse bias applied to a *p-n* junction is increased beyond a point, the junction breaks down and the reverse current rises sharply. The voltage at which breakdown happens is called the breakdown voltage and it depends on the width of the depletion region, which in turn depends on the doping level.

A normal *p-n* junction diode gets damaged at this point. Specially designed diodes like Zener diode can be operated at this region and can be used for the purpose of voltage regulation in circuits. There are two mechanisms that are responsible for breakdown under increasing reverse voltage.

i) Avalanche breakdown

Avalanche breakdown occurs in a lightly doped junctions which have wide depletion region. When reverse bias voltage exceeds a certain value, the minority charge carriers are accelerated by reverse voltage and their kinetic energy increases. These charge carriers collide

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 209 19-12-2022 16:38:55